

B3 — Atomic and Laser Physics

Trinity Term 2023 — Model Solutions

Question 1 — LS -coupling, terms, and the anomalous Zeeman effect in cadmium

Problem statement. (a) Explain LS -coupling and the conditions under which it applies, and list the spectroscopic terms arising from $3s^2$, $3s3p$, and $3s3d$. (b) Derive the Landé g -factor in LS -coupling. (c) Cadmium ($5s^2$ ground configuration) is placed in a magnetic flux density of 1.5 T; the Zeeman patterns of three lines $\lambda_A = 340.4$ nm, $\lambda_B = 346.8$ nm, $\lambda_C = 361.4$ nm are tabulated, with all three sharing a common upper and a common lower term. Deduce J and g_J for the levels of lines A and C , and propose L, S values for the two terms. (d) Identify line B . (e) List all allowed transitions between these two terms and compute the Zeeman pattern of one of them.

(a) The LS -coupling scheme and example terms

Condition: residual electrostatic repulsion dominates the spin-orbit interaction (light atoms, low Z , since spin-orbit $\propto Z^4$).

Couple orbital angular momenta first:

$$\mathbf{L} = \sum_i \ell_i, \quad \mathbf{S} = \sum_i \mathbf{s}_i. \quad (1)$$

Couple resultants to $\mathbf{J}$:

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \quad (2)$$

Good quantum numbers L, S, J, M_J ; terms labelled $^{2S+1}L_J$.

Terms from each configuration:

- $3s^2$: closed sub-shell with two equivalent electrons; Pauli forces $L = 0, S = 0$. Term: $\boxed{{}^1S_0}$.
- $3s3p$: $\ell_1 = 0, \ell_2 = 1 \Rightarrow L = 1; S = 0, 1$. Terms: $\boxed{{}^1P_1, {}^3P_{0,1,2}}$.
- $3s3d$: $\ell_1 = 0, \ell_2 = 2 \Rightarrow L = 2; S = 0, 1$. Terms: $\boxed{{}^1D_2, {}^3D_{1,2,3}}$.

(b) Derivation of the Landé g -factor

Zeeman Hamiltonian with $\mathbf{B} = B\hat{\mathbf{z}}$ and $g_s \simeq 2$:

$$\hat{H}_Z = \frac{\mu_B}{\hbar} \mathbf{B} \cdot (\hat{\mathbf{L}} + 2\hat{\mathbf{S}}). \quad (3)$$

Write $\hat{\mathbf{L}} + 2\hat{\mathbf{S}} = \hat{\mathbf{J}} + \hat{\mathbf{S}}$:

$$\hat{H}_Z = \frac{\mu_B}{\hbar} \mathbf{B} \cdot (\hat{\mathbf{J}} + \hat{\mathbf{S}}). \quad (4)$$

Project $\hat{\mathbf{S}}$ onto $\hat{\mathbf{J}}$ (perpendicular components average to zero):

$$\hat{H}_Z = g_J \frac{\mu_B}{\hbar} B \hat{J}_z, \quad (5)$$

with

$$g_J = 1 + \frac{\langle \hat{\mathbf{S}} \cdot \hat{\mathbf{J}} \rangle}{\langle \hat{\mathbf{J}}^2 \rangle}. \quad (6)$$

Expand $\hat{\mathbf{L}}^2 = (\hat{\mathbf{J}} - \hat{\mathbf{S}})^2$:

$$\hat{\mathbf{L}}^2 = \hat{\mathbf{J}}^2 + \hat{\mathbf{S}}^2 - 2\hat{\mathbf{S}} \cdot \hat{\mathbf{J}}. \quad (7)$$

Solve for $\hat{\mathbf{S}} \cdot \hat{\mathbf{J}}$:

$$\langle \hat{\mathbf{S}} \cdot \hat{\mathbf{J}} \rangle = \frac{1}{2} \hbar^2 [J(J+1) + S(S+1) - L(L+1)]. \quad (8)$$

Insert into (6) with $\langle \hat{\mathbf{J}}^2 \rangle = \hbar^2 J(J+1)$:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \quad (9)$$

Split $1 = 3/2 - J(J+1)/[2J(J+1)]$:

$$\boxed{g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}}. \quad (10)$$

(c) J and g_J for the levels involved in lines A and C

Compute the Zeeman unit at $B = 1.5$ T:

$$\frac{\mu_B B}{h} = 13.996 \text{ GHz/T} \times 1.5 \text{ T} = 20.99 \text{ GHz} \approx 21 \text{ GHz}. \quad (11)$$

Identify component shifts with $\Delta f / (\mu_B B / h) = g_u m_u - g_l m_l$, $\Delta m \in \{0, \pm 1\}$.

Line A (f_A , $f_A \pm 10.5$). Three symmetric components; spacing $10.5 \text{ GHz} = \frac{1}{2} \mu_B B / h$ is the signature of $J = 0 \leftrightarrow J = 1$ (one level does not split). Set spacing $g \mu_B B / h = 10.5 \text{ GHz}$:

$$g = \frac{1}{2}. \quad (12)$$

$$\boxed{\text{line } A : J = 0 \text{ (} g \text{ undefined), } J = 1 \text{ (} g = \frac{1}{2} \text{)}}. \quad (13)$$

Line C (9 components, f_C , ± 10.5 , ± 21 , ± 31.5 , ± 52.5). Largest shift in units of $\mu_B B / h$:

$$\pm 52.5 / 21 = \pm \frac{5}{2}. \quad (14)$$

Try $J_u = 2$, $J_l = 1$ with $g_l = 1/2$ (shared with line A). Extremal $(m_u, m_l) = (\pm 2, \pm 1)$:

$$|2g_u - g_l| = \frac{5}{2}. \quad (15)$$

Insert $g_l = 1/2$:

$$2g_u - \frac{1}{2} = \frac{5}{2}. \quad (16)$$

Solve for g_u :

$$g_u = \frac{3}{2}. \quad (17)$$

Tabulate $g_u m_u - g_l m_l$ for all $\Delta m \in \{0, \pm 1\}$:

(m_u, m_l)	$g_u m_u - g_l m_l$
$(\pm 2, \pm 1)$	$\pm 5/2$
$(\pm 1, \pm 1)$	± 1
$(\pm 1, 0)$	$\pm 3/2$
$(0, \pm 1)$	$\mp 1/2$
$(0, 0)$	0

(18)

The shift set $\{0, \pm 1/2, \pm 1, \pm 3/2, \pm 5/2\} \times \mu_B B/h$ matches the observed components.

line C : $J = 2$ ($g = \frac{3}{2}$), $J = 1$ ($g = \frac{1}{2}$).

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Identifying L, S . Apply (10) to the shared $J = 1, g = 1/2$ level:

$$\frac{3}{2} + \frac{S(S+1) - L(L+1)}{2 \cdot 1 \cdot 2} = \frac{1}{2}. \quad (20)$$

Solve for the bracket:

$$S(S+1) - L(L+1) = -4. \quad (21)$$

Try $L = 2, S = 1$:

$$2 - 6 = -4. \checkmark \quad (22)$$

Hence this term is 3D_1 . Apply (10) to the $J = 2, g = 3/2$ level:

$$\frac{3}{2} + \frac{S(S+1) - L(L+1)}{2 \cdot 2 \cdot 3} = \frac{3}{2}. \quad (23)$$

Hence:

$$S(S+1) - L(L+1) = 0. \quad (24)$$

Try $L = 1, S = 1$:

$$2 - 2 = 0. \checkmark \quad (25)$$

Hence this term is 3P_2 , with $J = 0$ partner 3P_0 on line A .

two terms: 3P ($L = 1, S = 1$) and 3D ($L = 2, S = 1$).

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(d) Identification of line B

Line B has six components but no central (π) component. Missing $m = 0 \rightarrow m' = 0$ is the signature of $J \rightarrow J$ with $J = J' \neq 0$. The only available $J = 1$ levels are 3P_1 and 3D_1 .

Compute $g(^3P_1)$ from (10) with $L = S = J = 1$:

$$g(^3P_1) = \frac{3}{2} + \frac{2-2}{4} = \frac{3}{2}. \quad (27)$$

Recall $g(^3D_1) = 1/2$ from part (c).

Take $\{g_u, g_l\} = \{3/2, 1/2\}$ and list $g_u m_u - g_l m_l$ with $\Delta m \in \{0, \pm 1\}$, $0 \rightarrow 0$ forbidden:

$$\{\pm \frac{1}{2}, \pm 1, \pm \frac{3}{2}\} \times \mu_B B/h = \{\pm 10.5, \pm 21, \pm 31.5\} \text{ GHz}. \quad (28)$$

Pattern matches the observed six components.

line B : $^3P_1 \leftrightarrow ^3D_1$.

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(e) All allowed ${}^3\text{P} \leftrightarrow {}^3\text{D}$ transitions and a worked pattern

Apply E1 selection rules $\Delta L = \pm 1$, $\Delta S = 0$, $\Delta J = 0, \pm 1$ (not $0 \rightarrow 0$). Allowed pairings:

$$\begin{array}{l|l} {}^3\text{P}_0 \leftrightarrow {}^3\text{D}_1 & \text{(line A)} \\ {}^3\text{P}_1 \leftrightarrow {}^3\text{D}_1 & \text{(line B)} \\ {}^3\text{P}_1 \leftrightarrow {}^3\text{D}_2 & \\ {}^3\text{P}_2 \leftrightarrow {}^3\text{D}_1 & \text{(line C)} \\ {}^3\text{P}_2 \leftrightarrow {}^3\text{D}_2 & \\ {}^3\text{P}_2 \leftrightarrow {}^3\text{D}_3 & \end{array} \quad (30)$$

Worked pattern: ${}^3\text{P}_1 \leftrightarrow {}^3\text{D}_2$. Compute $g({}^3\text{D}_2)$ from (10) with $L = 2, S = 1, J = 2$:

$$g({}^3\text{D}_2) = \frac{3}{2} + \frac{2-6}{12} = \frac{3}{2} - \frac{1}{3} = \frac{7}{6}. \quad (31)$$

Take $g_u = 7/6$ (${}^3\text{D}_2$), $g_l = 3/2 = 9/6$ (${}^3\text{P}_1$); tabulate $g_u m_u - g_l m_l$ for $\Delta m \in \{0, \pm 1\}$:

(m_u, m_l)	$g_u m_u - g_l m_l$ [$\mu_B B/h$]	Δf [GHz]
$(\pm 2, \pm 1)$	$\pm(\frac{14}{6} - \frac{9}{6}) = \pm\frac{5}{6}$	± 17.5
$(\pm 1, \pm 1)$	$\pm(\frac{7}{6} - \frac{9}{6}) = \mp\frac{1}{3}$	∓ 7
$(\pm 1, 0)$	$\pm\frac{7}{6}$	± 24.5
$(0, \pm 1)$	$\mp\frac{3}{2}$	∓ 31.5
$(0, 0)$	0	0

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Collect distinct shifts (taking f_D as the unperturbed line):

$$\boxed{f_D, f_D \pm 7, f_D \pm 17.5, f_D \pm 24.5, f_D \pm 31.5 \text{ GHz.}} \quad (33)$$

Question 2 — Multi-electron atoms, quantum defects, alkali spectra

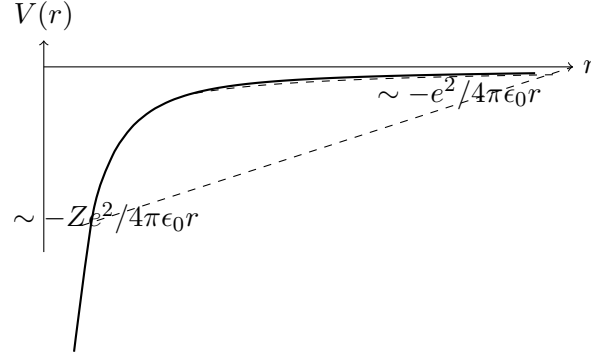
Problem statement. (a) State the approximations underlying the radial Schrödinger equation for the i -th electron of a multi-electron atom and sketch $V(r)$. (b) Identify the rubidium absorption series at $\{794.8, 780.0, 421.6, 420.2, 359.2, 358.7\}$ nm, compute the quantum defects δ_l , estimate the next doublet wavelength, and explain why the doublets close up. (c) Why is $\delta_{4d} = 1.23$ smaller than the δ_l found in (b), and is the 4d level more or less bound than 6s? (d) Identify the spontaneous-emission lines from the $\lambda = 421.6$ nm excitation. (e) Krypton's ground configuration; estimate the energy needed to excite a 5d electron.

(a) Approximations and the effective potential

Approximations underlying the radial equation:

1. **Central field:** replace electron–electron repulsion by an angle-averaged spherical mean field $V(r_i)$.
2. **Independent electron:** total wavefunction is a Slater determinant of single-electron orbitals.
3. **Separation of variables:** $\psi = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$, substitute $P_{n\ell} = rR_{n\ell}$ to obtain the 1D radial equation with centrifugal term $\hbar^2 \ell(\ell+1)/(2mr^2)$.
4. **Self-consistency:** iterate $V(r)$ from the orbitals (Hartree–Fock).

Limits: $V(r) \rightarrow -Ze^2/4\pi\epsilon_0 r$ as $r \rightarrow 0$ (bare nucleus); $V(r) \rightarrow -e^2/4\pi\epsilon_0 r$ as $r \rightarrow \infty$ (fully screened):



For $\ell > 0$ the centrifugal barrier excludes the electron from the core: high- ℓ orbitals are hydrogenic.

(b) Identifying the Rb absorption series

Rubidium: single $5s_{1/2}$ valence electron above a Kr-like core. E1 selection rule $\Delta\ell = \pm 1$ permits only the principal series $5s \rightarrow np_{1/2,3/2}$; six lines form three fine-structure doublets for $n = 5, 6, 7$.

Quantum-defect formula:

$$E_{n\ell} = -\frac{R_\infty}{(n - \delta_\ell)^2}, \quad R_\infty = 13.606 \text{ eV}. \quad (34)$$

Apply to the $5s$ ground state with $|E_{5s}| = 4.177 \text{ eV}$:

$$n_{5s}^* = \sqrt{\frac{13.606}{4.177}} = 1.804. \quad (35)$$

Hence:

$$\delta_{5s} = 5 - 1.804 = 3.20. \quad (36)$$

Procedure for each line: $|E_{np}| = |E_{5s}| - hc/\lambda$, $n^* = \sqrt{R_\infty/|E_{np}|}$, $\delta = n - n^*$. Use $hc = 1239.84 \text{ eV nm}$:

λ/nm	$hc/\lambda/\text{eV}$	$ E_{np} /\text{eV}$	n^*	n	δ_{np}
794.8	1.5599	2.6171	2.280	5	2.720
780.0	1.5895	2.5875	2.293	5	2.707
421.6	2.9408	1.2362	3.318	6	2.682
420.2	2.9505	1.2265	3.331	6	2.669
359.2	3.4517	0.7253	4.331	7	2.669
358.7	3.4566	0.7204	4.346	7	2.654

(37)

Identify the longer- λ (more bound) member as $np_{1/2}$, with $\delta \simeq 2.71$; the shorter- λ member is $np_{3/2}$, with $\delta \simeq 2.66$. The defect is roughly n -independent.

Next doublet ($n = 8$). Take a mean $\delta_{8p} \simeq 2.66$:

$$(8 - 2.66)^2 = 5.34^2 = 28.52. \quad (38)$$

Compute the binding energy:

$$|E_{8p}| = \frac{13.606}{28.52} = 0.477 \text{ eV}. \quad (39)$$

Transition energy:

$$\Delta E = 4.177 - 0.477 = 3.700 \text{ eV}. \quad (40)$$

Mean wavelength:

$$\bar{\lambda}_{8p} = \frac{1239.84}{3.700} = 335.1 \text{ nm}.$$

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Why doublets close up. Spin-orbit splitting scales as $\langle r^{-3} \rangle_{np} \propto (n - \delta)^{-3}$:

$$\Delta E_{\text{FS}}(np) \propto \frac{1}{(n - \delta)^3}. \quad (42)$$

Observed splittings 29.6 meV (5p), 9.7 meV (6p), 4.8 meV (7p) follow this scaling.

(c) Quantum defect of the 4d level; comparison with 6s binding energy

Centrifugal barrier $\propto \ell(\ell + 1)/r^2$ excludes high- ℓ orbitals from the core, leaving them in the screened single-charge tail and pushing $\delta_\ell \rightarrow 0$. Hence $\delta_{4d} = 1.23$ is much smaller than $\delta_{5s} = 3.20$ and $\delta_{np} \simeq 2.7$.

Compute $|E_{4d}|$ from (34):

$$(4 - 1.23)^2 = 2.77^2 = 7.673. \quad (43)$$

$$|E_{4d}| = \frac{13.606}{7.673} = 1.773 \text{ eV}. \quad (44)$$

Compute $|E_{6s}|$ with $\delta_{6s} \simeq \delta_{5s} = 3.20$:

$$(6 - 3.20)^2 = 2.80^2 = 7.840. \quad (45)$$

$$|E_{6s}| = \frac{13.606}{7.840} = 1.736 \text{ eV}. \quad (46)$$

Compare:

$$\boxed{|E_{4d}| > |E_{6s}| : 4d \text{ is (slightly) more tightly bound than } 6s.} \quad (47)$$

(d) Decay channels from the level reached by $\lambda = 421.6 \text{ nm}$

Identify the pump $\lambda = 421.6 \text{ nm}$ as the larger-defect member of the $5s \rightarrow 6p$ doublet, hence $5s_{1/2} \rightarrow 6p_{1/2}$. Apply $\Delta\ell = \pm 1$, $\Delta J = 0, \pm 1$ (not $0 \rightarrow 0$): $6p_{1/2}$ can emit to $ns_{1/2}$ ($\Delta\ell = -1$) and to $nd_{3/2}$ ($\Delta\ell = +1$; $nd_{5/2}$ excluded by $\Delta J = 2$).

Use the binding energies from (b) and (c):

$$|E_{5p_{1/2}}| = 2.617 \text{ eV}, \quad |E_{5p_{3/2}}| = 2.588 \text{ eV}, \quad (48)$$

$$|E_{6p_{1/2}}| = 1.236 \text{ eV}, \quad |E_{6s_{1/2}}| = 1.736 \text{ eV}, \quad |E_{4d_{3/2}}| = 1.773 \text{ eV}. \quad (49)$$

(The empirical $6s_{1/2}$ value is closer to 1.680 eV; constant- δ estimate suffices for line identification.)

Compute $6p_{1/2} \rightarrow 6s_{1/2}$ with empirical $|E_{6s}| = 1.680 \text{ eV}$:

$$\Delta E = 1.680 - 1.236 = 0.444 \text{ eV}. \quad (50)$$

$$\lambda = \frac{1239.84}{0.444} = 2791 \text{ nm}. \quad (51)$$

$$\boxed{2791.3 \text{ nm} : 6p_{1/2} \rightarrow 6s_{1/2}.} \quad (52)$$

Compute $6p_{1/2} \rightarrow 4d_{3/2}$:

$$\Delta E = 1.773 - 1.236 = 0.537 \text{ eV}. \quad (53)$$

$$\lambda = \frac{1239.84}{0.537} = 2308 \text{ nm}. \quad (54)$$

$$\boxed{2293.9 \text{ nm} : 6p_{1/2} \rightarrow 4d_{3/2}.} \quad (55)$$

Cascade $6s_{1/2} \rightarrow 5p_{1/2}$ (with $|E_{6s}| = 1.680 \text{ eV}$):

$$\Delta E = 2.617 - 1.680 = 0.937 \text{ eV.} \quad (56)$$

$$\lambda = \frac{1239.84}{0.937} = 1323 \text{ nm.} \quad (57)$$

Cascade $6s_{1/2} \rightarrow 5p_{3/2}$:

$$\Delta E = 2.588 - 1.680 = 0.908 \text{ eV.} \quad (58)$$

$$\lambda = \frac{1239.84}{0.908} = 1366 \text{ nm.} \quad (59)$$

$$\boxed{1323.5 \text{ nm} : 6s_{1/2} \rightarrow 5p_{1/2}; \quad 1366.5 \text{ nm} : 6s_{1/2} \rightarrow 5p_{3/2}.} \quad (60)$$

Cascade $4d_{3/2} \rightarrow 5p_{1/2}$ ($\Delta J = -1$):

$$\Delta E = 2.617 - 1.773 = 0.844 \text{ eV.} \quad (61)$$

$$\lambda = \frac{1239.84}{0.844} = 1469 \text{ nm.} \quad (62)$$

Cascade $4d_{3/2} \rightarrow 5p_{3/2}$ ($\Delta J = 0$):

$$\Delta E = 2.588 - 1.773 = 0.815 \text{ eV.} \quad (63)$$

$$\lambda = \frac{1239.84}{0.815} = 1521 \text{ nm.} \quad (64)$$

$$\boxed{1475.2 \text{ nm} : 4d_{3/2} \rightarrow 5p_{1/2}; \quad 1528.8 \text{ nm} : 4d_{3/2} \rightarrow 5p_{3/2}.} \quad (65)$$

Final D lines $5p_J \rightarrow 5s_{1/2}$:

$$\boxed{780.0 \text{ nm} : 5p_{3/2} \rightarrow 5s_{1/2}; \quad 794.8 \text{ nm} : 5p_{1/2} \rightarrow 5s_{1/2}.} \quad (66)$$

Direct return:

$$\boxed{421.6 \text{ nm} : 6p_{1/2} \rightarrow 5s_{1/2}.} \quad (67)$$

All listed transitions satisfy $\Delta \ell = \pm 1$ and $\Delta J \in \{0, \pm 1\}$ (no $0 \rightarrow 0$).

(e) Krypton: ground configuration and 5d-excitation energy

Fill shells up to $Z = 36$:

$$\boxed{\text{Kr} : [\text{Ar}] 3d^{10} 4s^2 4p^6.} \quad (68)$$

Promote a 4p electron: lowest 5d-bearing configuration is $4p^5 5d$. The 5d electron sees a Kr^+ core (net charge $+e$); $\ell = 2$ is non-penetrating, so $\delta_{5d} \approx 0$ and the binding is hydrogenic.

Apply the hydrogenic formula with $n = 5$:

$$|E_{5d}| \simeq \frac{R_\infty}{n^2} = \frac{13.606}{25} = 0.544 \text{ eV.} \quad (69)$$

Excitation = ionisation cost minus 5d binding:

$$\Delta E \simeq E_i - |E_{5d}| = 14.0 - 0.544 \text{ eV.} \quad (70)$$

$$\boxed{\Delta E_{4p^6 \rightarrow 4p^5 5d} \simeq 13.5 \text{ eV.}} \quad (71)$$

Question 3 — Two- and three-level dynamics, Rabi flopping, and two-photon transitions

Problem statement. A two-level atom with eigenstates $|1\rangle, |2\rangle$ and energies $E_1 < E_2$ is driven by $\hat{V}(t) = e\mathcal{E} \cos \omega t$, with general time-dependent state $\Psi = c_1 \psi_1 e^{-iE_1 t/\hbar} + c_2 \psi_2 e^{-iE_2 t/\hbar}$. (a) Derive $\dot{c}_2$ in the form quoted, defining the Rabi frequency. (b) Solve in the rotating-wave approximation (RWA) for $|c_2(t)|^2$ when $c_1 \simeq 1$. (c) For a three-level Λ -like system driven near a two-photon resonance with $|\Delta| \gg |\delta|$, derive $|c_3(t)|^2 = (W^2/\beta^2) \sin^2(\beta t)$ and compare with the one-photon result. (d) Justify the simplification of the more general coupling matrix.

(a) Coupled equations and Rabi frequency

Substitute $\Psi = c_1 \psi_1 e^{-iE_1 t/\hbar} + c_2 \psi_2 e^{-iE_2 t/\hbar}$ into $i\hbar \dot{\Psi} = (\hat{H}_0 + \hat{V})\Psi$ and use $\hat{H}_0 \psi_j = E_j \psi_j$ to cancel the E_j terms:

$$i\hbar [\dot{c}_1 \psi_1 e^{-iE_1 t/\hbar} + \dot{c}_2 \psi_2 e^{-iE_2 t/\hbar}] = \hat{V}(t) [c_1 \psi_1 e^{-iE_1 t/\hbar} + c_2 \psi_2 e^{-iE_2 t/\hbar}]. \quad (72)$$

Project onto ψ_2^* using $\langle \psi_i | \psi_j \rangle = \delta_{ij}$:

$$i\hbar \dot{c}_2 e^{-iE_2 t/\hbar} = c_1 \langle 2 | \hat{V} | 1 \rangle e^{-iE_1 t/\hbar} + c_2 \langle 2 | \hat{V} | 2 \rangle e^{-iE_2 t/\hbar}. \quad (73)$$

Drop $\langle 2 | \hat{x} | 2 \rangle = 0$ by parity:

$$i\hbar \dot{c}_2 e^{-iE_2 t/\hbar} = c_1 e\mathcal{E} \cos \omega t \langle 2 | \hat{x} | 1 \rangle e^{-iE_1 t/\hbar}. \quad (74)$$

Define $\omega_{21} = (E_2 - E_1)/\hbar$ and the Rabi frequency:

$$\boxed{\Omega \equiv \frac{e\mathcal{E}}{\hbar} \langle 2 | \hat{x} | 1 \rangle.} \quad (75)$$

Divide by $i\hbar e^{-iE_2 t/\hbar}$:

$$\boxed{\dot{c}_2(t) = -i\Omega \cos(\omega t) e^{i\omega_{21} t} c_1(t).} \quad (76)$$

(b) Weak excitation in the RWA

Set $c_1 \simeq 1$, $c_2(0) = 0$; integrate (76):

$$c_2(t) = -i\Omega \int_0^t \cos(\omega t') e^{i\omega_{21} t'} dt'. \quad (77)$$

Expand $\cos \omega t' = (e^{i\omega t'} + e^{-i\omega t'})/2$:

$$c_2(t) = -\frac{i\Omega}{2} \int_0^t [e^{i(\omega_{21} + \omega)t'} + e^{i(\omega_{21} - \omega)t'}] dt'. \quad (78)$$

Drop the fast $e^{i(\omega_{21} + \omega)t'}$ term (RWA):

$$c_2(t) \simeq -\frac{i\Omega}{2} \int_0^t e^{i(\omega_{21} - \omega)t'} dt'. \quad (79)$$

Evaluate the integral:

$$c_2(t) \simeq -\frac{i\Omega}{2} \frac{e^{i(\omega_{21} - \omega)t} - 1}{i(\omega_{21} - \omega)}. \quad (80)$$

Simplify:

$$c_2(t) \simeq -\frac{\Omega}{2} \frac{e^{i(\omega_{21}-\omega)t} - 1}{\omega_{21} - \omega}. \quad (81)$$

Apply $e^{ix} - 1 = 2ie^{ix/2} \sin(x/2)$:

$$c_2(t) \simeq -i\Omega \frac{e^{i(\omega_{21}-\omega)t/2} \sin[\frac{1}{2}(\omega_{21} - \omega)t]}{\omega_{21} - \omega}. \quad (82)$$

Take the modulus squared:

$$|c_2(t)|^2 = \frac{\Omega^2}{(\omega - \omega_{21})^2} \sin^2\left[\frac{1}{2}(\omega - \omega_{21})t\right]. \quad (83)$$

On resonance ($\omega = \omega_{21}$) the result $\rightarrow \Omega^2 t^2 / 4$.

(c) Two-photon transition $1 \rightarrow 3$ via off-resonant level 2

Apply RWA to the $\dot{c}_2$ equation, keeping the slow phase $\omega_{21} - \omega_a = -\Delta$:

$$\dot{c}_2 \simeq -\frac{i\Omega_a}{2} e^{-i\Delta t} c_1. \quad (84)$$

For $\dot{c}_3$, identify the slow phase. Use $\omega_{23} = \omega_{21} - \omega_{31}$ and the given $\omega_a - \omega_b = \omega_{31} + \delta$, i.e. $\omega_b = \omega_a - \omega_{31} - \delta$:

$$-\omega_{23} + \omega_b = -(\omega_{21} - \omega_{31}) + (\omega_a - \omega_{31} - \delta) = \omega_a - \omega_{21} - \delta. \quad (85)$$

Simplify:

$$-\omega_{23} + \omega_b = \Delta - \delta. \quad (86)$$

Hence:

$$\dot{c}_3 \simeq -\frac{i\Omega_b}{2} e^{i(\Delta-\delta)t} c_2. \quad (87)$$

Adiabatic elimination of level 2. Ansatz $c_2 = A(t)e^{-i\Delta t}$ with $|\dot{A}| \ll |\Delta A|$ (Rabi rates $\ll |\Delta|$). Substitute into (84) with $c_1 \simeq 1$:

$$\dot{A} e^{-i\Delta t} - i\Delta A e^{-i\Delta t} = -\frac{i\Omega_a}{2} e^{-i\Delta t}. \quad (88)$$

Drop $\dot{A}$ and cancel $e^{-i\Delta t}$:

$$-i\Delta A = -\frac{i\Omega_a}{2}. \quad (89)$$

Solve for A :

$$A = \frac{\Omega_a}{2\Delta}. \quad (90)$$

$$c_2(t) \simeq \frac{\Omega_a}{2\Delta} e^{-i\Delta t}. \quad (91)$$

Effective two-photon coupling. Substitute (91) into (87):

$$\dot{c}_3 \simeq -\frac{i\Omega_b}{2} e^{i(\Delta-\delta)t} \cdot \frac{\Omega_a}{2\Delta} e^{-i\Delta t}. \quad (92)$$

Combine exponentials:

$$\dot{c}_3 \simeq -iW e^{-i\delta t}, \quad \boxed{W \equiv \frac{\Omega_a \Omega_b}{4\Delta}}. \quad (93)$$

Integrate with $c_3(0) = 0$:

$$c_3(t) = -iW \frac{e^{-i\delta t} - 1}{-i\delta}. \quad (94)$$

Simplify:

$$c_3(t) = \frac{W}{\delta} (e^{-i\delta t} - 1). \quad (95)$$

Apply $e^{-i\delta t} - 1 = -2ie^{-i\delta t/2} \sin(\delta t/2)$:

$$c_3(t) = -\frac{2iW}{\delta} e^{-i\delta t/2} \sin(\delta t/2). \quad (96)$$

Take $|c_3|^2$:

$$|c_3(t)|^2 = \frac{4W^2}{\delta^2} \sin^2(\delta t/2). \quad (97)$$

Identify $\beta = \delta/2$:

$$\boxed{|c_3(t)|^2 = \frac{W^2}{\beta^2} \sin^2(\beta t), \quad \beta = \frac{\delta}{2}, \quad W = \frac{\Omega_a \Omega_b}{4\Delta}}. \quad (98)$$

Same form as (83): an effective two-level Rabi oscillation $1 \leftrightarrow 3$ with coupling W and detuning $\delta/2$. The $1/\Delta$ suppression of W is the price of using a far-detuned intermediate level to suppress spontaneous loss.

(d) Why the simplified equations are a good approximation

Expand each cos and identify the slow phase of every coupling:

- $\Omega_a \cos(\omega_a t) e^{i\omega_{21}t}$: slow phase $\omega_{21} - \omega_a = -\Delta$. **Kept.**
- $\Phi_b \cos(\omega_b t) e^{i\omega_{21}t}$: slow phase $\omega_{21} - \omega_b \simeq \omega_{31} + \Delta$ (using $\omega_b = \omega_a - \omega_{31} - \delta$). **Drop** (fast).
- $\Phi_a \cos(\omega_a t) e^{-i\omega_{23}t}$: slow phase $-\omega_{23} + \omega_a = \Delta + \omega_{31}$. **Drop** (fast).
- $\Omega_b \cos(\omega_b t) e^{-i\omega_{23}t}$: slow phase $-\omega_{23} + \omega_b = \Delta - \delta$. **Kept.**

Dropped terms oscillate at frequencies $\sim \omega_{31}$, much faster than the dynamics they drive ($\sim \Omega$). Their cumulative contribution is suppressed by $\Omega/\omega_{31} \ll 1$. Keeping only the two resonant pairings (laser a on $1 \leftrightarrow 2$, laser b on $2 \leftrightarrow 3$) is the multi-frequency RWA.

Question 4 — Rate equations, gain saturation, and Lamb dip in a He–Ne laser

Problem statement. A homogeneously broadened transition between levels of degeneracies g_1, g_2 has populations N_1, N_2 , with the upper level pumped at rate R_2 and decaying only by spontaneous emission A_{21} , while the lower level has lifetime τ_1 . (a) Steady-state rate equations and inversion. (b) Add a narrow-band radiation field; derive the gain-saturation form $\alpha = \alpha_0/(1 + I/I_{\text{sat}})$ and identify α_0, I_{sat} . (c) Sketch the gain seen by a weak probe in three different irradiation conditions. (d) HeNe Lamb dip: identify two broadening mechanisms, estimate physical parameters from the curves. (e) Explain the dip and its broadening with discharge current.

(a) Rate equations and steady-state inversion

Write rate equations:

$$\dot{N}_2 = R_2 - A_{21}N_2, \quad (99)$$

$$\dot{N}_1 = A_{21}N_2 - N_1/\tau_1. \quad (100)$$

Set $\dot{N}_2 = 0$:

$$N_2 = \frac{R_2}{A_{21}}. \quad (101)$$

Set $\dot{N}_1 = 0$:

$$N_1 = \tau_1 A_{21} N_2 = R_2 \tau_1. \quad (102)$$

Form the (degeneracy-weighted) inversion:

$$N^* \equiv N_2 - \frac{g_2}{g_1} N_1 = \frac{R_2}{A_{21}} - \frac{g_2}{g_1} R_2 \tau_1. \quad (103)$$

Factor R_2 :

$$\boxed{N^* = R_2 \left(\frac{1}{A_{21}} - \frac{g_2}{g_1} \tau_1 \right)}. \quad (104)$$

Positive inversion requires $1/A_{21} > (g_2/g_1)\tau_1$ (the lower-level bottleneck condition).

(b) Saturated gain coefficient

Let W_{21} be the stimulated downward rate; the upward rate is $(g_2/g_1)W_{21}$. Add stimulated terms to (99)–(100):

$$\dot{N}_2 = R_2 - A_{21}N_2 - W_{21} \left(N_2 - \frac{g_2}{g_1} N_1 \right), \quad (105)$$

$$\dot{N}_1 = A_{21}N_2 - N_1/\tau_1 + W_{21} \left(N_2 - \frac{g_2}{g_1} N_1 \right). \quad (106)$$

Add (105) and (106) and set both LHS to zero:

$$0 = R_2 - N_1/\tau_1. \quad (107)$$

Hence $N_1 = R_2 \tau_1$, unchanged. From (105) in steady state:

$$R_2 = A_{21}N_2 + W_{21}N^*. \quad (108)$$

Solve for N_2 :

$$N_2 = \frac{R_2 - W_{21}N^*}{A_{21}}. \quad (109)$$

Substitute into $N^* = N_2 - (g_2/g_1)N_1$:

$$N^* = \frac{R_2 - W_{21}N^*}{A_{21}} - \frac{g_2}{g_1}R_2\tau_1. \quad (110)$$

Collect N^* terms:

$$N^* \left(1 + \frac{W_{21}}{A_{21}} \right) = R_2 \left(\frac{1}{A_{21}} - \frac{g_2}{g_1}\tau_1 \right) = N_0^*. \quad (111)$$

Solve for the saturated inversion:

$$N^* = \frac{N_0^*}{1 + W_{21}/A_{21}}. \quad (112)$$

Use the cross-section to relate W_{21} and I at the laser frequency:

$$W_{21} = \frac{\sigma(\omega) I}{\hbar\omega}, \quad \sigma(\omega) = \frac{\pi^2 c^2}{\omega^2} A_{21} g_H(\omega). \quad (113)$$

Multiply N^* by $\sigma(\omega)$ to get the gain coefficient $\alpha = \sigma(\omega)N^*$:

$$\alpha(\omega) = \frac{\sigma(\omega)N_0^*}{1 + W_{21}/A_{21}}. \quad (114)$$

At line centre ($\omega = \omega_{21}$):

$$\boxed{\alpha(\omega_{21}) = \frac{\alpha_0}{1 + I/I_{\text{sat}}}}, \quad (115)$$

with

$$\alpha_0 = \sigma(\omega_{21}) N_0^* = \frac{\pi^2 c^2}{\omega_{21}^2} A_{21} g_H(\omega_{21}) R_2 \left(\frac{1}{A_{21}} - \frac{g_2}{g_1}\tau_1 \right), \quad (116)$$

and, equating $W_{21}/A_{21} = I/I_{\text{sat}}$,

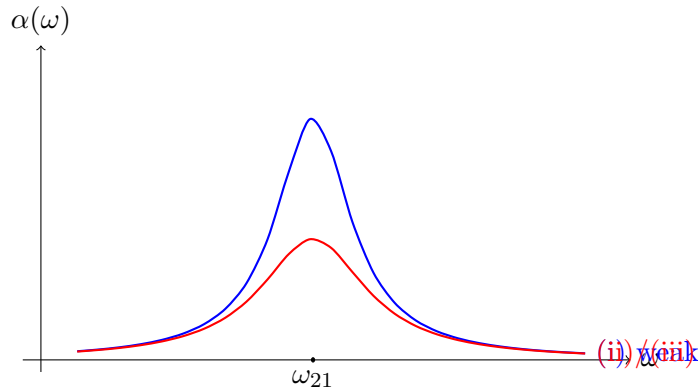
$$I_{\text{sat}} = \frac{\hbar\omega_{21}A_{21}}{\sigma(\omega_{21})}. \quad (117)$$

Substitute $\sigma(\omega_{21})$:

$$\boxed{I_{\text{sat}} = \frac{\hbar\omega_{21}^3}{\pi^2 c^2 g_H(\omega_{21})}}. \quad (118)$$

At $I = I_{\text{sat}}$ the inversion drops to half its small-signal value.

(c) Sketches of gain vs. frequency



- (i) Weak probe sees the unsaturated Lorentzian, peak α_0 , FWHM $\Delta\omega_H$.
- (ii) Single saturating beam at $I = I_{\text{sat}}$. Use (115) with local Lorentzian $L(\omega)$:

$$\alpha(\omega) = \frac{\alpha_0 L(\omega)}{1 + IL(\omega)/I_{\text{sat}}}, \quad L(\omega) = \frac{1}{1 + 4(\omega - \omega_{21})^2/\Delta\omega_H^2}. \quad (119)$$

Evaluate at line centre ($L = 1$, $I = I_{\text{sat}}$):

$$\alpha(\omega_{21}) = \alpha_0/2. \quad (120)$$

Solve $\alpha(\omega) = \alpha_0/4$ for FWHM: $L = 1/3 \Rightarrow \text{FWHM} = \sqrt{2} \Delta\omega_H$ (power broadening).

(iii) Homogeneous medium: every atom contributes at every frequency, so the strong pump at ω_{21} reduces N^* uniformly. The weak probe sees the same curve as (ii) — no hole is burned. A spectral hole would arise only in an inhomogeneously broadened medium.

(d) Two broadening mechanisms in the He–Ne laser at 633 nm

Doppler (inhomogeneous). Each atom emits at its Doppler-shifted frequency; the resulting Gaussian envelope has FWHM

$$\Delta\nu_D = \frac{2}{\lambda} \sqrt{\frac{2k_B T \ln 2}{m}}. \quad (121)$$

Take the observed gain envelope width as $\Delta\nu_D \simeq 1.5$ GHz (the dip-bearing region of the curve is roughly half the full envelope). Solve (121) for T :

$$T = \frac{mc^2}{8k_B \ln 2} \left(\frac{\Delta\nu_D}{\nu} \right)^2. \quad (122)$$

Insert $m = 20 \text{ u} = 3.32 \times 10^{-26} \text{ kg}$, $\nu = c/\lambda = 4.74 \times 10^{14} \text{ Hz}$, $\Delta\nu_D = 1.5 \text{ GHz}$:

$$T \simeq 400 \text{ K}, \quad (123)$$

a reasonable discharge-tube temperature.

Natural + collisional (homogeneous). The Lorentzian FWHM is

$$\Delta\nu_H = \frac{1}{2\pi\tau_2} + \frac{1}{2\pi\tau_1} + \frac{n\sigma_{\text{coll}}\bar{v}}{2\pi}. \quad (124)$$

Use $\tau_2 \sim 30 \text{ ns}$, $\tau_1 \sim 10 \text{ ns}$:

$$\Delta\nu_N = \frac{1}{2\pi\tau_2} + \frac{1}{2\pi\tau_1} \sim 20 \text{ MHz}. \quad (125)$$

At pressure $\sim 1 \text{ mbar}$ ($n \sim 2.5 \times 10^{22} \text{ m}^{-3}$), with $\sigma_{\text{coll}} \sim 10^{-19} \text{ m}^2$, $\bar{v} \sim 700 \text{ m s}^{-1}$:

$$\Delta\nu_{\text{coll}} = \frac{n\sigma_{\text{coll}}\bar{v}}{2\pi} \sim 10 \text{ MHz}. \quad (126)$$

Total $\Delta\nu_H \sim 20 \text{ MHz}$ to 30 MHz , consistent with the 30 MHz Lamb-dip width of curve A.

(e) Why a Lamb dip appears, and why it broadens with R_2

Mechanism. A cavity standing wave at frequency ω is the sum of two counter-propagating running waves. Each saturates atoms whose longitudinal velocity satisfies $\omega(1 \mp v_z/c) = \omega_{21}$:

$$v_z = \pm c(\omega - \omega_{21})/\omega_{21}. \quad (127)$$

For $\omega \neq \omega_{21}$ the two classes are different atoms (no mutual saturation). For $\omega = \omega_{21}$ both reduce to $v_z = 0$ (same atoms), doubling the saturation rate and depressing the gain — the Lamb dip. Its width is the homogeneous width $\Delta\nu_H$.

Broadening with R_2 . Higher discharge current $\rightarrow$ larger $R_2 \rightarrow$ larger intracavity intensity I . The hole at $v_z = 0$ power-broadens via the $\sqrt{1 + I/I_{\text{sat}}}$ factor of (c):

$$\boxed{\Delta\nu_{\text{dip}} = \Delta\nu_H \sqrt{1 + I/I_{\text{sat}}}.} \quad (128)$$

Apply to curves $A \rightarrow C$ ($30 \rightarrow 40$ MHz):

$$\frac{I}{I_{\text{sat}}} = \left(\frac{40}{30}\right)^2 - 1 \simeq 0.78. \quad (129)$$

Power broadening is the dominant mechanism; collisional broadening at higher current makes a smaller contribution.